

Part D

Project Overview

The data, code, models, experiments, and supporting documentation for the Data Science Capstone project carried out for STADIOcare Group are all included in this repository.

The project's main goal is to determine whether patient demand, hospital bed utilization, and length-of-stay patterns can be predicted using historical patient, operational, and workforce data.

The intended business application will help STADIOcare achieve its goal of becoming a more data-led organization by supporting more proactive capacity and workforce planning.

Repository Structure

STADIOcare-Capstone/

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|— README.md

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|— data/

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| |— raw/

|

| |— processed/

|

| |— README.md

|

|— models/

|

| |— trained/

|

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|— experiments/

|

| |— setup/

|

| |— results/

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| |— README.md

|

|— scripts/

```
| ├── statistical/
| ├── comparison/
| └── visualisation/
|
| ├── notebooks/
|
| ├── documentation/
| ├── motivation/
| ├── problem-statement/
| └── data-request/
|
| ├── requirements.txt
|
| └── .gitignore
```

Repository Contents

Datasets

The data/ directory contains datasets used during the project.

- data/raw/ contains original datasets received from the client.
- data/processed/ contains cleaned and transformed datasets prepared for analysis and modelling.

Client-sensitive or confidential data should not be uploaded to a public repository.

Models

The models/ directory contains trained models and associated model artefacts.

The models/trained/ folder will contain the final models selected for the project, together with relevant model documentation where required.

Experimental Setup

The experiments/setup/ directory contains information required to reproduce the experiments, including:

- Model configurations
- Hyperparameters

- Feature-selection decisions
- Dataset versions
- Experimental assumptions
- Evaluation methodology

Experimental Results

The experiments/results/ directory contains the outputs generated from experiments and model evaluations.

These may include:

- Performance metrics
- Model comparison results
- Prediction results
- Evaluation tables
- Experiment summaries

Statistical Helper and Comparison Scripts

The scripts/statistical/ directory contains statistical analysis and helper scripts.

The scripts/comparison/ directory contains scripts used to compare models, approaches and experimental results.

Visualisation Scripts

The scripts/visualisation/ directory contains code used to generate project visualisations, including exploratory data analysis charts, model evaluation plots and other analytical visualisations.

Notebooks

The notebooks/ directory contains Jupyter notebooks used for exploratory data analysis, data preparation, modelling and evaluation.

Notebooks should be clearly named and numbered where appropriate to indicate their purpose and sequence.

Documentation

The documentation/ directory contains supporting project documentation.

This includes:

- Project motivation
- Problem statement

- Client data request
- Other relevant Capstone documentation

Data Privacy

STADIOcare is a healthcare organisation and project data may contain sensitive information.

No personally identifiable information or confidential client data should be committed to a public GitHub repository.

Where necessary, data should be de-identified, anonymised or excluded from the repository in accordance with the client's requirements and applicable data governance rules.

Reproducibility

The repository is structured to support reproducibility of the Data Science project.

Scripts, notebooks, experimental configurations and model results should be version controlled where appropriate. Dataset versions and important assumptions should also be documented.

Project Status

The repository will be updated throughout the Capstone project as additional datasets, analysis, models, experiments and results become available.